

Is Twitter for the Birds?

Estimating the Causal Effect of Twitter (X) Sentiment on Stock Returns

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Abstract

This paper estimates the causal effect of daily Twitter sentiment on intraday stock returns using two identification strategies: firm fixed-effects OLS and Double Machine Learning (DoubleML) with XGBoost nuisance learning. The primary dataset is a Bloomberg daily panel of 503 S&P 500 firms spanning January 2025 to March 2026 (approximately 160,000 firm-day observations), extended to a 1.19-million-observation S&P 500 panel and a 1.97-million-observation Russell 3000 panel over 2016–2026 for temporal and cross-sectional analysis. In the S&P 500 sample, Twitter sentiment is statistically insignificant in all fixed-effects specifications; the DoubleML estimate is -0.924 ($p < 0.01$), but a placebo test—regressing yesterday’s return on today’s sentiment—yields a coefficient of $+2.348$ ($p < 0.01$), indicating substantial reverse causality from prices to sentiment. News sentiment is the more robust predictor across both frameworks (-0.759^{***} in DoubleML). A Fama-MacBeth replication of Gu and Kurov (2020) produces insignificant coefficients on lagged Twitter sentiment in every calendar year from 2016 to 2025 for the S&P 500, but significant effects in the broader Russell 3000. Cap-group decomposition reveals a sharp gradient: the

Panel OLS sentiment coefficient is +0.375*** for large-cap firms, rising to +0.804*** for mid-cap and +0.942*** for small-cap firms, consistent with an information-environment explanation in which Twitter provides marginal predictive content primarily where institutional analyst coverage is sparse. Daily long-short portfolios sorted on sentiment generate positive Sharpe ratios in most years, suggesting the signal is concentrated in the tails of the cross-sectional distribution rather than in a linear average slope.

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1 Introduction

Does social media sentiment causally affect intraday stock price movements, and does the answer depend on the type of firm? This paper provides evidence for both questions using three Bloomberg panels: a daily S&P 500 panel over 2025–2026 with a robust set of confounders; an extended panel of the S&P 500 from 2016–2026; and an extended Russell 3000 panel over 2016–2026. These are used to employ firm fixed-effects OLS, Double Machine Learning (DoubleML), and Fama-MacBeth cross-sectional regressions to estimate the effect of Twitter/X sentiment on stock returns across the majority of the US market.

The question matters for three core reasons. Social media sentiment has been widely studied, but as its proliferation has increased, its value as a news source has degraded. The ability to measure its influence in the market provides a meaningful indicator of its usefulness as an information sharing tool. Secondly, as retail investor numbers have grown, the number of investors with informal access to market data has also grown. These traders are more likely to use platforms such as Twitter, Reddit, and StockTwits for information than formal, expensive platforms such as Bloomberg. This investor segment is also more active in the small and mid-cap sections of the market. Lastly, the question of Twitter’s causal influence remains open. While much literature documents that text-derived sentiment correlates with subsequent returns, correlation is not causation. Firms that attract more social media attention may differ systematically from those that do not; sentiment itself may simply respond to price changes rather than cause them. Resolving this requires identification strategies capable of isolating the sentiment effect from confounders and from reverse causality. Investigating these topics provides useful information on how information diffuses through the market and why.

Other research has shown that the fraction of negative words in the Wall Street Journal *Abreast of the Market* column predicts next-day Dow Jones returns and trading volume, with the effect reversing within a week. In a study by Gu and Kurov (2020) Fama-MacBeth regressions have been used to show that lagged Twitter sentiment predicts open-to-open

stock returns in a 2015–2017 sample, with the effect concentrated at a one-day lag (a study replicated in this paper for both robustness and trend analysis). These studies provide the empirical background for this analysis. This paper goes a bit further by examining whether effects are heterogeneous across the market-capitalization distribution and by substantially testing for reverse causality. Based on that gap, I sought to make three contributions.

The first is methodological. By applying the PLR model of Chernozhukov et al. (2018), combined with XGBoost as a nuisance learner, I obtain an estimate of the sentiment effect that is asymptotically unconfounded by high-dimensional covariates, regardless of the functional form of their relationship with the treatment and outcome. The DoubleML estimator achieves this by cross-fitting: nuisance models for both $E[\text{twitter_sent} \mid X]$ and $E[\text{return} \mid X]$ are trained on held-out folds, so the final coefficient is not contaminated by in-sample overfitting. This is, to my knowledge, the first application of DoubleML to intraday Twitter sentiment and stock returns.

The second contribution is the systematic documentation of a reverse causality concern. I pair the primary DoubleML estimates with a firm-level placebo test that regresses yesterday’s return on today’s Twitter sentiment. Under the null of no reverse causation, this coefficient should be indistinguishable from zero. It is not. The estimated coefficient is +2.348 ($p < 0.01$), substantially larger in magnitude than any of the forward-looking estimates. This result is inconsistent with clean causal transmission from sentiment to prices and raises the possibility that Bloomberg’s sentiment measure largely reflects the market’s own prior-day behavior rather than an independent information signal.

The third contribution is the cap-group analysis of the Russell 3000. By extending the Fama-MacBeth and Panel OLS specifications to approximately 1.97 million firm-day observations across the full Russell 3000 and partitioning firms by time-averaged market capitalization, I document a sharp split in sentiment effect: +0.375*** for large-cap firms (market cap >\$10B); +0.804*** for mid-cap firms (\$2B–\$10B); and +0.942*** for small-cap firms (<\$2B).

An interaction specification also confirms that the effect is linearly decreasing with firm size ($\text{twitter_sent} \times \log_mkt_cap = -0.148^*$) across the entire 11 year panel, indicating that the smaller the firm, the stronger Twitter sentiment’s influence. This also informs the discrepancy between the null result on the S&P 500 and the significant effects documented in other studies that include smaller firms, indicating that Twitter sentiment is most informative precisely where institutional information is sparsest.

The main findings can be summarized as follows. In the S&P 500 firm fixed-effects specifications ($N \approx 160,000$ firm-day observations), `twitter_sent` is statistically insignificant across all four treatment constructions. News sentiment enters with a coefficient of -1.056 ($p < 0.01$) and is the dominant predictor in the linear OLS framework. The DoubleML estimate for `twitter_sent` is -0.924 ($p < 0.01$), but the placebo coefficient of $+2.348$ makes causal attribution very weak. The Fama-MacBeth coefficient on lagged Twitter sentiment is insignificant in every calendar year from 2016 through 2025 for the S&P 500 universe, with no structural break or temporal trend. For large-cap S&P 500 constituents, Twitter sentiment provides no statistically reliable predictive signal in any year examined.

The Russell 3000 results stand in contrast. The full-sample Panel OLS coefficient on `twitter_sent` is $+0.708$ ($p < 0.01$) across 1.97 million firm-day observations, positive and significant where the S&P 500 estimate was negative and insignificant. The Fama-MacBeth series by cap group shows consistent significance for mid- and small-cap segments across most years, with the strongest effects concentrated in 2018–2020. A daily long-short strategy—long top-decile, short bottom-decile sentiment firms within each cap group—generates positive returns in most years, suggesting that even where the average linear coefficient is modest, Twitter sentiment produces exploitable cross-sectional dispersion in realized returns. This last result is interesting however, because this strategy was also shown to be effective for the S&P 500.

The remainder of the paper proceeds as follows. Section 2 reviews the relevant literature on

sentiment and asset pricing, social media as an information channel, and causal identification in high-frequency panels. Section 3 describes the Bloomberg panels, variable construction, the Russell 3000 data infrastructure, and sample properties. Section 4 presents the FE-OLS, DoubleML, Fama-MacBeth, and cap-group specifications with full notation and identifying assumptions. Section 5 reports point estimates, robustness checks, the placebo and lag-structure tests, and the full Russell 3000 cap-group analysis. Section 6 interprets the central tension between the DoubleML estimates and the reverse causality evidence, and the cap-group gradient in the context of the information environment hypothesis. Section 7 concludes with limitations and directions for future work, including a custom entity-level sentiment tool that may isolate causally relevant signal from the Bloomberg composite.

2 Literature Review

This section reviews three strands of literature that inform the present study: (1) investor sentiment and asset pricing, (2) social media as a financial information channel, and (3) causal identification in high-frequency panel settings. The paper sits at their intersection and contributes a formal causal framework using double debiased machine learning to a domain that has traditionally relied on linear models.

2.1 Media Sentiment and Asset Pricing

The theoretical foundations for a sentiment channel in asset prices relies on the concept that media content drives investor behavior as new information is acquired. As this information spreads through the market, it influences trader behavior through bid pricing, selling, and buying.

Early text-based evidence comes from Tetlock (2007), who found that media pessimism in the Wall Street Journal *Abreast of the Market* column predicts next-day downward pressure on the Dow Jones Industrial Average, with the effect fully reversing over the following week. This

demonstrates how information available influences trader behavior. Social media sentiment is just another form of such information.

2.2 Social Media as a Financial Information Channel

A second branch of literature examines social media specifically as a source of price-relevant information transmission, distinct from traditional news outlets. Antweiler and Frank (2004) analyzed 1.5 million messages posted on Yahoo Finance and Raging Bull message boards and found that message volume predicts next-day trading volume and volatility, with a more modest link to returns. Their work establishes the concept that user-generated financial discussion contains informational content, but leaves open whether the effect is economically large or causal.

This topic was explicitly covered in a paper that is the most direct predecessor to this study. Gu and Kurov (2020) used firm-level daily Twitter sentiment from a commercial data provider to predict open-to-open stock returns in a Fama-MacBeth, pooled OLS framework over January 2015 to February 2017. Their key results are as follows: (i) a significant positive coefficient on one-day lagged Twitter sentiment (+0.136***); (ii) a no-reversal test showing only lag 1 is significant when lags 1–5 are entered jointly; and (iii) a profitable long-short strategy with an annualized Sharpe ratio of 3.17. These findings are interpreted as evidence that Twitter sentiment conveys firm-specific information that is absorbed within one trading day.

The Gu and Kurov (2020) paper also provides an example of the Bloomberg Twitter sentiment analysis tool used in formal research. This paper replicates the Gu and Kurov (2020) specifications on an extended panel of both the SP 500 and the Russell 3000 from 2016 through 2026 to both test for the robustness of the initial study, and to examine sentiment sensitivity based on firm size as measured in market capitalization.

2.3 Contribution to Literature

The present paper seeks to expand the understanding of how social media sentiment influences the market. It takes the research question and variable constructions from the social media–finance literature of Antweiler and Frank (2004) and Gu and Kurov (2020) and applies more modern causal identification strategies. This contributes to the broader debate about whether investor sentiment has real pricing effects. Three features distinguish it from prior work:

1. **Formal causal framework.** The DoubleML PLR estimator relaxes linearity assumptions that constrain FE and Fama-MacBeth OLS estimates, providing a coefficient that is interpretable as a causal partial effect under the assumption that the confounder vector of confounders X_{it} is sufficiently exogenous.
2. **Reverse causality diagnostics.** Rather than assuming the direction of causation, the paper explicitly tests for it through a placebo regression and lag-structure analysis. The finding that the reverse coefficient (+2.348) exceeds the forward estimate (−0.924) in absolute value disciplines the causal interpretation throughout.
3. **Updated sample period.** Prior Twitter-returns studies concentrate on 2011–2017, a period of rapid social media adoption. The present sample (2024–2025) captures a more mature informational environment in which algorithmic trading, bot activity, and information saturation may have attenuated the retail sentiment channel. The temporal comparison with Gu and Kurov (2020) is informative about the stability of the Twitter effect over time.

3 Data

3.1 Source and Sample Construction

All data are sourced from the Bloomberg Terminal via the Bloomberg Desktop API (`blpapi`). I construct three panel datasets serving distinct roles in the analysis:

1. **Primary S&P 500 panel.** Daily observations from January 1, 2025 through March 25, 2026, comprising 160,487 firm-day observations across 503 S&P 500 constituents. This panel contains the full set of treatment, outcome, and control variables used in the FE-OLS and DoubleML specifications.
2. **Extended S&P 500 panel.** Daily observations from January 1, 2016 through February 27, 2026, comprising approximately 1.19 million firm-day observations across the same 503 firms. This panel contains a reduced variable set (prices, volume, market capitalization, sentiment scores, and Fama-French factors) and is used exclusively for the temporal Fama-MacBeth analysis that tracks the sentiment coefficient from 2016 to 2026.
3. **Russell 3000 panel.** Daily observations from January 1, 2016 through February 27, 2026, comprising approximately 1.97 million firm-day observations across the Russell 3000 constituents. This panel is used for the cap-group decomposition analysis and contains prices, volume, market capitalization, and sentiment scores. The Russell 3000 analysis employs the same Fama-MacBeth and Panel OLS specifications as the S&P 500 analysis, enabling a direct universe comparison.

Non-trading days (weekends, holidays) are excluded by dropping observations where both opening and closing prices are missing. Bloomberg-specific missing value codes (`#N/A N/A`, `#N/A Field Not Applicable`) are recoded as `NA` prior to any numerical transformation.

3.2 Variable Definitions

Table 1 defines all variables used in the analysis. The S&P 500 primary panel uses the full variable set; the extended S&P 500 and Russell 3000 panels use the subset marked with a dagger ($\dagger$). Those where the “Bloomberg Field” is blank, reflect variables computed by combining values from the other data fields.

Table 1: Variable Definitions

Variable	Bloomberg Field	Definition	Panel
<code>return</code>	—	$\text{px_close}_{it} - \text{px_open}_{it}$ (open-to-close, \$)	Primary
<code>return_oo</code>	—	Open-to-open return (%)	Extended
<code>return_oo_adj</code>	—	FF4-factor residual of <code>return_oo</code>	Extended
<code>twitter_sent</code>	TWITTER_SENTIMENT_DAILY_AVG	Daily avg. Twitter sentiment ($[-1, 1]$)	All
<code>twitter_neg_count</code>	TWITTER_NEG_SENTIMENT_COUNT	Count of negative-polarity tweets	All
<code>news_sent</code>	NEWS_SENTIMENT_DAILY_AVG	Daily avg. news sentiment ($[-1, 1]$)	All
<code>px_high</code>	PX_HIGH	Intraday high price (\$)	Primary
<code>px_low</code>	PX_LOW	Intraday low price (\$)	Primary
<code>mkt_cap</code>	CUR_MKT_CAP	Market cap (millions, \$)	All
<code>log_mkt_cap</code>	—	$\log(\text{mkt_cap})$	Extended
<code>total_equity</code>	TOTAL_EQUITY	Book equity (millions, \$)	Primary
<code>debt_to_equity</code>	TOT_DEBT_TO_TOT_EQY	Total debt / total equity (%)	Primary
<code>volume</code>	PX_VOLUME	Daily share volume (shares)	All
<code>abnorm_vol</code>	—	% deviation of volume from ticker mean	Extended
<code>vol_rs</code>	—	Rogers-Satchell realized volatility	Extended
<code>rsi_30</code>	RSI_30D	30-day Relative Strength Index	Primary
<code>ma_50</code>	MOV_AVG_50D	50-day moving average of price (\$)	Primary
<code>bid_ask_spread</code>	AVERAGE_BID_ASK_SPREAD	Daily avg. bid-ask spread (\$)	All
<code>lagk</code>	—	$\text{return}_{i,t-k}$, $k \in \{1, 2, 3, 5, 7\}$	Primary
<code>cap_group</code>	—	Large / Mid / Small (see Section 3.3)	Russ

Table 1: Variable Definitions

Variable	Bloomberg Field	Definition	Panel
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The primary outcome for the S&P 500 specifications, **return**, is the raw dollar price change from market open to close on day t for firm i . This differs from the percentage return used in Gu and Kurov (2020); the dollar denomination preserves interpretability of treatment coefficients as dollar-per-unit sentiment effects, and firm fixed effects absorb cross-sectional differences in share price levels. For the extended and Russell panels, the outcome is the factor-adjusted open-to-open return (**return_oo_adj**), which is the appropriate dependent variable for the Fama-MacBeth specifications following Gu and Kurov (2020).

The primary treatment variable, **twitter_sent**, is Bloomberg’s proprietary daily average Twitter sentiment score. Bloomberg applies natural language processing to the universe of public tweets mentioning a given firm’s name, ticker, or cashtag, assigning each tweet a score on $[-1, +1]$ and averaging across all tweets within the calendar day. The variable **twitter_neg_count** counts only those tweets classified with negative polarity, following Teti et al. (2019). The news sentiment variable, **news_sent**, is constructed analogously from Bloomberg-sourced news articles.

3.3 Cap-Group Classification

For the Russell 3000 analysis, firms are assigned to one of three market-capitalization groups based on their time-averaged market capitalization over the full sample period:

- **Large cap:** $\text{mkt_cap}_i > \$10\text{B}$ ($N = 763,274$ firm-days)
- **Mid cap:** $\$2\text{B} \leq \text{mkt_cap}_i \leq \10B ($N = 640,932$ firm-days)
- **Small cap:** $\text{mkt_cap}_i < \$2\text{B}$ ($N = 568,068$ firm-days)

Group membership is fixed for the duration of the sample using the time-average rather than daily market capitalization, so that firms do not migrate between groups due to short-run price volatility. This ensures that within-group temporal comparisons are not confounded by within-firm capitalization shifts.

The large-cap group corresponds to an expanded version of the S&P 500 universe, with the mid-cap and small-cap groups to the Russell 2000. The thresholds of \$2B and \$10B are standard in institutional investment practice and are consistent with the Bloomberg index classification methodology.

3.4 Measurement Limitations

Three features of the Bloomberg sentiment data warrant discussion.

Proprietary classification. Bloomberg does not disclose the dictionary, model architecture, or training data underlying its sentiment classification. This precludes replication with alternative NLP tools and prevents diagnostic checks on whether specific tweet categories (e.g., bot-generated, reply threads, retweets) are included or filtered. The Loughran and McDonald (2011) critique of general-purpose sentiment dictionaries applies with unknown force: it is possible that Bloomberg’s classifier misattributes domain-specific financial language.

Daily Aggregation. The sentiment scores are daily averages. Bloomberg aggregates all tweets within a 24-hour window without distinguishing pre-market, intraday, and after-hours activity. Because the primary outcome variable is measured from market open to market close, a tweet posted after the close is counted in the same day’s sentiment as one posted before the open. This aggregation mismatch biases the design against detecting clean causal effects and may introduce reverse causality if after-hours tweets respond to the day’s price action that has already been recorded in **return**. This concern is directly assessed by the placebo test in Section 5.4.

Selection in sentiment coverage. Not all firm-days have Twitter sentiment scores. Missing

sentiment days are not randomly distributed across firms or time: large-cap, high-volume firms receive disproportionate coverage. In the S&P 500 extended panel, Twitter sentiment is non-missing for 908,781 of 1,190,286 firm-days (76% coverage). Coverage is sparser for Russell 3000 firms, particularly for the small-cap segment where lower public visibility reduces the daily tweet count below Bloomberg’s classification threshold on many days. Summary statistics for all three panels are reported in Tables 2–4 and document the extent of missingness across key variables.

3.5 Sample Properties

S&P 500 primary panel. The panel spans 503 firms over approximately 320 trading days (January 2025–March 2026), yielding 160,487 firm-day observations with non-missing price data. The effective sample size for the full FE-OLS specification (Model 2) drops to 147,128 observations after listwise deletion of rows with missing controls, primarily `total_equity` and `debt_to_equity` for newly listed firms. The baseline specification (Model 1) retains 160,103 observations. For DoubleML, XGBoost handles missing covariates natively; observations are excluded only when the treatment variable itself is missing. Summary statistics are reported in Table 2. Twitter sentiment is non-missing for 160,103 of 160,487 firm-days (99.8% coverage); the high coverage reflects the S&P 500’s large-cap composition, which generates sufficient daily tweet volume to meet Bloomberg’s classification threshold on nearly every trading day.

Extended S&P 500 panel. The panel covers 2,633 unique trading days across the same 503 firms (January 2016–February 2026), yielding approximately 1.19 million firm-day observations. Twitter sentiment coverage begins in earnest around 2017 for most firms, with sporadic coverage in 2016. The Fama-MacBeth regressions exclude dates on which fewer than 30 firms report non-missing sentiment. The sample is truncated at February 28, 2026, the last available date in the Kenneth French factor library, to ensure clean four-factor risk adjustment throughout. Twitter sentiment is non-missing for 908,781 of 1,190,286 firm-days

(76% coverage). Summary statistics are reported in Table 3.

Russell 3000 panel. The panel covers approximately 2,921 firms across the 2016–2026 window, yielding 1,972,274 firm-day observations with non-missing price data. After restricting to firm-days with non-missing `twitter_sent`, the effective sample for Fama-MacBeth estimation is smaller (2,183,113 firm-days with non-missing sentiment across the full raw file); Fama-MacBeth regressions exclude dates on which fewer than 30 firms per cap group report non-missing sentiment. The large-cap group ($N = 763,274$) has the highest sentiment coverage given its overlap with the S&P 500 universe; the small-cap group ($N = 568,068$) has the lowest, as lower public visibility reduces daily tweet counts below Bloomberg’s classification threshold on many days. Summary statistics are reported in Table 4.

Table 2: Summary Statistics — Primary S&P 500 Panel (2025–2026)

Variable	Mean	SD	Min	Max	N
<i>Outcome and treatment variables</i>					
<code>return</code> (\$)	0.061	8.810	−582.570	521.270	160,487
<code>twitter_sent</code>	0.020	0.118	−0.999	0.992	160,103
<code>twitter_neg_count</code>	2.044	25.721	0.000	3,121.000	159,232
<code>news_sent</code>	−0.002	0.179	−1.000	0.998	160,487
<i>Price and market variables</i>					
<code>px_high</code> (\$)	225.256	473.577	7.560	8,567.560	160,487
<code>px_low</code> (\$)	219.948	463.754	6.860	8,472.000	160,487
<code>mkt_cap</code> (\$M)	119,619	373,869	3,430	5,031,072	160,475
<code>volume</code> (shares)	1,831,016	4,059,166	2,789	214,110,528	160,487
<i>Firm fundamentals</i>					
<code>total_equity</code> (\$M)	23,824	54,630	−14,231	719,703	157,412
<code>debt_to_equity</code> (%)	164.870	522.343	0.000	9,191.429	148,633
<i>Technical indicators</i>					
<code>rsi_30</code>	50.912	8.516	16.704	87.570	160,257
<code>ma_50</code> (\$)	221.816	472.937	8.329	8,946.917	160,097

Notes: Daily firm-level panel. 503 S&P 500 constituents, January 2025 through March 2026 (approximately 320 trading days). `return` = open-to-close price change (\$). `twitter_sent` = Bloomberg daily average Twitter sentiment $\in [-1, 1]$. `twitter_neg_count` = count of negative-polarity tweets. `news_sent` = Bloomberg daily average news sentiment $\in [-1, 1]$. `mkt_cap` = market capitalisation (\$M). `rsi_30` = 30-day Relative Strength Index. N reflects non-missing observations per variable; listwise deletion reduces the FE-OLS full model (Model 2) to 147,128 observations.

Table 3: Summary Statistics — Full Panel (2016–2026)

Variable	Mean	SD	Min	Max	<i>N</i>
return_oo	0.0665	2.1386	-47.7455	107.7465	1,190,286
twitter_sent	0.0257	0.1651	-1.0000	1.0000	908,781
news_sent	0.0108	0.3019	-1.0000	1.0000	935,145
vol_rs	0.0292	0.1008	0.0000	21.2587	1,190,789
abnorm_vol	-0.0000	84.8711	-99.9982	15263.2174	1,190,785
log_mkt_cap	10.3627	1.1180	6.2183	15.4311	1,190,699
bid_ask_spread	0.1987	5.9468	0.0003	3209.5924	1,190,731
return_oo_adj	-0.0000	2.1321	-47.9895	107.4738	1,156,294

Notes: Daily firm-level panel. return_oo = open-to-open return (%). vol_rs = Rogers-Satchell realized volatility. abnorm_vol = abnormal volume (% deviation from ticker mean). log_mkt_cap = log market capitalization. return_oo_adj = Fama-French-Carhart 4-factor residual.

Table 4: Summary Statistics — Russell 3000 Panel (2016–2026)

Variable	Mean	SD	Min	Max	<i>N</i>
<i>Sentiment variables</i>					
twitter_sent	0.027	0.165	-1.000	1.000	2,183,113
news_sent	0.019	0.262	-1.000	1.000	2,169,011
<i>Market and price variables</i>					
mkt_cap (\$M)	17,064	98,558	0.371	5,031,072	5,015,692
log_mkt_cap	7.775	1.883	-0.992	15.431	5,015,692
volume (shares)	578,573	3,198,797	0	837,455,680	5,019,920
bid_ask_spread (\$)	0.602	13.577	0.000	999.991	5,000,337
<i>Cap-group breakdown (firm-day observations)</i>					
Large cap ($\overline{\text{mkt_cap}} > \10B)		—			763,274
Mid cap ($\$2\text{B} - \10B)		—			640,932
Small cap ($\overline{\text{mkt_cap}} < \2B)		—			568,068

Notes: Daily firm-level panel. Approximately 2,921 Russell 3000 constituents, January 2016 through February 2026 (truncated at the last available Fama-French factor date). The analysis dependent variable (`return_oo_adj`) is the Fama-French-Carhart four-factor residual of the open-to-open return, computed at runtime; 1,156,294 firm-days have a non-missing factor-adjusted return. Cap-group *N* reflects firm-days used in the Panel OLS cap-group regressions (Table 14), restricted to firms with non-missing price data. `mkt_cap` denotes each firm’s time-averaged market capitalisation used for fixed group assignment. `twitter_sent` coverage is lower than total firm-days because Bloomberg’s classifier requires a minimum tweet volume threshold.

4 Methodology

This section describes the two primary identification strategies—firm fixed-effects OLS and Double Machine Learning—as well as the Fama-MacBeth procedure used for temporal and cross-sectional analysis, the cap-group classification, and the long-short portfolio construction.

4.1 Firm Fixed-Effects OLS

The baseline identification strategy is a within-firm panel regression. Let Y_{it} denote the open-to-close price change (`return`) for firm i on day t , D_{it} the sentiment treatment, and X_{it} a vector of controls. The estimating equation is:

$$Y_{it} = \alpha_i + \beta D_{it} + \gamma' X_{it} + \varepsilon_{it},$$

where α_i is a firm fixed effect and ε_{it} is an idiosyncratic error. The parameter of interest is β , the within-firm partial effect of sentiment on same-day return after conditioning on observable time-varying confounders.

Firm fixed effects absorb all time-invariant, firm-specific factors that could jointly influence sentiment coverage and price levels—including industry membership, market capitalization quintile, analyst following, and the firm’s typical trading clientele. The identifying variation is purely within-firm and over time. This is appropriate because Bloomberg’s Twitter sentiment measure has markedly heterogeneous coverage across firms (Section 3.4): the fixed effect controls for the fact that firms with high average sentiment may simply be large, liquid, or widely-followed, and thus differ from low-coverage firms on dimensions that also affect returns.

The control vector X_{it} in the full specification (Model 2) includes: intraday price range (`px_high`, `px_low`), market capitalization (`mkt_cap`), book equity (`total_equity`), leverage (`debt_to_equity`), trading volume (`volume`), news sentiment (`news_sent`), negative tweet

count (`twitter_neg_count`), 30-day RSI (`rsi_30`), and the 50-day moving average (`ma_50`). Standard errors are heteroskedasticity-robust throughout.

Four treatment constructions are estimated. Model 1 uses the continuous daily average sentiment score $D_{it} = \text{twitter_sent} \in [-1, +1]$. Model 2 augments Model 1 with the full control vector. Model 3 replaces the treatment with a negative-clipped measure, $D_{it} = \min(\text{twitter_sent}_{it}, 0)$, which equals zero on positive-sentiment days and the raw score on negative-sentiment days—isolating whether the return effect is asymmetric in the negative tail. Model 4, following Teti et al. (2019), uses daily negative tweet counts (`twitter_neg_count`) as the treatment. All four models include firm fixed effects.

Limitation. The FE-OLS specification assumes that the confounder–outcome and confounder–treatment relationships are both linear. Nonlinear interactions among controls—for example, between momentum (RSI) and sentiment during high-volatility regimes—will contaminate the β estimate if the true data-generating process is non-additive. The DoubleML specification in Section 4.2 relaxes this assumption.

4.2 Double Machine Learning

The primary causal estimator is the Partially Linear Regression (PLR) model of Chernozhukov et al. (2018), estimated via the `DoubleML` Python package. The PLR model is:

$$\begin{aligned} Y_{it} &= \theta_0 D_{it} + g_0(X_{it}) + U_{it}, & E[U_{it} \mid D_{it}, X_{it}] &= 0, \\ D_{it} &= m_0(X_{it}) + V_{it}, & E[V_{it} \mid X_{it}] &= 0, \end{aligned}$$

where θ_0 is the causal parameter of interest, and $g_0(\cdot)$ and $m_0(\cdot)$ are unknown nuisance functions capturing the conditional expectations of the outcome and treatment given the full confounder vector X_{it} , respectively. Critically, g_0 and m_0 are estimated nonparametrically—here using XGBoost as the nuisance learner—so no linearity or additive separability is assumed for the confounders.

Cross-fitting. To prevent overfitting bias from contaminating the treatment effect estimate, I use $K = 5$ folds of cross-fitting: the nuisance functions $\hat{g}$ and $\hat{m}$ are trained on four folds and evaluated on the held-out fold, so the residuals $\hat{U}_{it}$ and $\hat{V}_{it}$ are always out-of-sample predictions. The final coefficient $\hat{\theta}$ is obtained by regressing $\hat{U}_{it}$ on $\hat{V}_{it}$ —the partialled-out outcome on the partialled-out treatment—aggregated across folds. The primary treatment-effect estimates use a single cross-fitting draw; standard errors are computed from the `DoubleML` package’s default asymptotic variance formula. For the lag-decay and impact-persistence robustness specifications (`lag_cates.py`), I repeat the full cross-fitting procedure over 20 random fold assignments and derive confidence intervals from 1,000 bootstrap draws, which stabilizes coefficient estimates against fold-specific randomness in the smaller effective samples obtained when conditioning on lagged or lead return windows.

XGBoost configuration. The nuisance learner is XGBoost (`XGBRegressor`), a gradient-boosted tree ensemble. Gradient boosting is well-suited to this setting because it handles mixed-type regressors, captures nonlinear and interaction effects implicitly through tree splits, and is robust to the collinearity among sentiment, momentum, and volume variables that can destabilize linear nuisance models. The same confounder set used in the FE-OLS full model is supplied to both nuisance learners.

Treatment variants. Three treatment series are estimated separately in the `DoubleML` framework: (1) `twitter_sent` (continuous); (2) `twitter_neg_count` (count of negative-polarity tweets, following Teti et al. (2019)); and (3) `news_sent` (continuous news sentiment). A fourth specification estimates each treatment against `px_high` (intraday price high) as the outcome rather than `return`, to test whether sentiment’s effect on the intraday price ceiling differs from its effect on the close-to-open spread.

Identifying assumption. Under the PLR, θ_0 is identified if $E[U_{it} \mid D_{it}, X_{it}] = 0$ —that is, if the confounder vector X_{it} fully controls for the non-random assignment of sentiment to firms and days. This is a conditional unconfoundedness assumption. It does not require an

instrument. It does require that no omitted variable jointly drives both `twitter_sent` and `return` after conditioning on the observable controls. The plausibility of this assumption is assessed through the placebo test described in Section 4.6.

4.3 Fama-MacBeth Procedure and Temporal Analysis

For the replication and temporal analysis, I follow Fama and MacBeth (1973) and Gu and Kurov (2020) in estimating daily cross-sectional regressions and aggregating the time series of coefficients. On each trading day t , I estimate:

$$r_{it} = a_t + b_t \cdot \text{twitter_sent}_{i,t-1} + \mathbf{c}_t' X_{it} + e_{it},$$

where r_{it} is the Fama-French-Carhart four-factor-adjusted return (`return_oo_adj`) for firm i on day t , and the treatment is the one-day lagged sentiment score. The confounder vector X_{it} includes five lags each of `return_oo_adj`, `abnorm_vol` (abnormal volume), `vol_rs` (Rogers-Satchell realized volatility), `log_mkt_cap`, and `bid_ask_spread`. The Fama-MacBeth estimate of the sentiment effect is $\bar{b} = T^{-1} \sum_{t=1}^T \hat{b}_t$, with Newey-West standard errors computed over the time series of $\hat{b}_t$ using a bandwidth of five lags to correct for serial correlation. Days on which fewer than 30 firms report non-missing sentiment are excluded from the aggregation.

Risk adjustment. Factor-adjusted returns (`return_oo_adj`) are constructed by estimating full-sample Fama-French-Carhart four-factor betas for each firm and subtracting the fitted four-factor component from the raw open-to-open return. Factor data are obtained from Kenneth French’s data library; the sample is truncated at the last available factor date (February 28, 2026) to avoid extrapolation. The residual `return_oo_adj` captures idiosyncratic price variation that cannot be attributed to market, size, value, or momentum exposures, and is the appropriate dependent variable for tests of sentiment-specific information.

Temporal windows. The Fama-MacBeth procedure is applied separately over annual,

quarterly, and monthly windows to characterize how the sentiment coefficient has evolved from 2016 to 2026. Results are reported graphically as coefficient-and-confidence-band plots and, for the annual cadence, in a tabular format. Monthly estimates are less reliable due to small cross-sections in some months and are used for visualization only.

4.4 Cap-Group Classification and Russell 3000 Analysis

To test whether the sentiment effect is heterogeneous across the market-capitalization spectrum, I classify all Russell 3000 firms into three groups based on their time-averaged market capitalization:

- **Large cap:** $\overline{\text{mkt_cap}}_i > \10B
- **Mid cap:** $\$2\text{B} \leq \overline{\text{mkt_cap}}_i \leq \10B
- **Small cap:** $\overline{\text{mkt_cap}}_i < \2B

I use time-averaged rather than daily market capitalization to assign group membership, so that firms do not migrate between groups due to price volatility within the sample window. This ensures that the cap-group comparison is stable and that within-group temporal analysis is not confounded by within-firm cap migration.

The Russell 3000 analysis proceeds in three stages. First, I estimate a pooled Panel OLS with firm fixed effects that includes a `twitter_sent` $\times$ `log_mkt_cap` interaction term, testing whether the sentiment coefficient is linearly decreasing in firm size across the full cross-section. Second, I run the Fama-MacBeth procedure separately for each cap group and plot the annual coefficient series. Third, I compute a daily long-short portfolio within each cap group to assess the economic magnitude of the cross-sectional dispersion in sentiment.

4.5 Long-Short Portfolio Construction

To assess the economic significance of the sentiment-return relationship, I construct a daily long-short portfolio following Gu and Kurov (2020). On each trading day t , all firms with non-missing `twitter_sent` are ranked into deciles. The strategy takes a long position in firms in the top decile (highest sentiment) and a short position in firms in the bottom decile (lowest sentiment), holding for 24 hours on an open-to-open basis. Portfolio returns are the equal-weighted average return across firms in each leg. For the cap-group analysis, deciles are formed separately within each cap group, so the strategy is self-contained per segment and not dominated by large-cap firms with superior sentiment coverage.

Performance is summarized by annualized Sharpe ratio (assuming 252 trading days), annualized return, and win rate (fraction of days on which the long leg outperforms the short leg). All reported figures are before transaction costs, and the strategy should be interpreted as an upper bound on achievable risk-adjusted performance.

4.6 Robustness Tests

Four robustness checks discipline the causal interpretation of the primary estimates.

Placebo test. Yesterday’s return (`lag1`) is regressed on today’s Twitter sentiment within firms. Under the null of no reverse causality, the coefficient should be indistinguishable from zero. A large, significant coefficient indicates that today’s sentiment responds to yesterday’s price movement—feedback from prices to social media—and constitutes evidence against a clean forward causal interpretation of the primary estimates.

No-reversal test. I enter five sentiment lags jointly (`twitter_senti,t-1` through `twitter_senti,t-7`, at lags 1, 2, 3, 5, 7) and test whether only lag 1 is significant. Under a causal interpretation in which information is absorbed within one trading day, the multi-lag regression should show significance concentrated at lag 1 with coefficients at longer lags near zero.

Lag decay. Each sentiment lag is estimated separately in a univariate DoubleML specification with the same confounder set as the primary model. If the coefficient is large at lag 1 and decays monotonically to zero by lag 5 or lag 7, this is consistent with a transient but real sentiment effect. A pattern of large, roughly equal coefficients at all lags is inconsistent with causal transmission through prices and more consistent with a common driver of both sentiment and returns.

Impact persistence. Current-day sentiment is regressed on future returns at leads +1 through +7 (same intervals as lag decay). Under the hypothesis that positive sentiment today drives positive returns tomorrow, the lead-1 coefficient should be positive and significant, with the effect decaying at longer leads. A flat or sign-inconsistent pattern across leads is evidence against a durable predictive signal.

5 Results

This section reports results in five stages: (1) FE-OLS estimates for the S&P 500 panel, (2) DoubleML PLR estimates for sentiment and news treatments, (3) the intraday price high specification, (4) robustness checks including the placebo test and lag structure analysis, and (5) the Russell 3000 cap-group analysis.

5.1 Fixed-Effects OLS: S&P 500 Panel

Table 5 reports the four FE-OLS specifications. The primary result is unambiguous: `twitter_sent` is statistically insignificant in all four models, and its coefficient does not change appreciably when the full confounder matrix is added.

Model 1, the baseline regression of `return` on `twitter_sent` with firm fixed effects only, produces a coefficient of -0.370 ($SE = 0.230$, $p > 0.10$). The R^2 of 0.002 confirms that Twitter sentiment alone explains essentially none of the within-firm variation in intraday returns. Model 2 adds the full control vector; the `twitter_sent` coefficient moves to -0.422

(SE = 0.325) and remains insignificant. The R^2 rises to 0.019, driven primarily by two significant controls: news sentiment (`news_sent` = -1.056 , $p < 0.01$) and the 30-day RSI (`rsi_30` = $+0.084$, $p < 0.01$). The negative news sentiment coefficient indicates that a one-unit increase in the Bloomberg daily news sentiment score is associated with a -1.06 cent same-day return within firms, a substantially larger and more reliably estimated effect than the Twitter channel. The positive RSI coefficient is consistent with short-term price momentum: firms on an upward 30-day trajectory tend to continue gaining within the trading day.

Models 3 and 4 test alternative treatment constructions. The negative-clipped measure ($\min(\text{twitter_sent}, 0)$, Model 3) produces a coefficient of -0.072 (SE = 0.271), smaller in absolute value than the unrestricted measure in Model 1, suggesting that the negative tail of Twitter sentiment is not a disproportionately powerful predictor relative to the full sentiment distribution. Model 4, using `twitter_neg_count` following Teti et al. (2019), yields a coefficient of $+0.000533$ (SE = 0.000971), not significant. The positive sign is consistent with the hypothesis that high tweet volume—even predominantly negative volume—signals elevated attention rather than bearish information, but the effect is economically trivial and statistically indistinguishable from zero.

5.2 DoubleML Estimates: Twitter and News Sentiment

Table 7 and Table 8 report the DoubleML PLR estimates. The contrast with the FE-OLS results is substantial.

Twitter sentiment. The PLR estimate for `twitter_sent` is -0.924 (SE = 0.226, $p < 0.01$), statistically significant at the 1% level. This coefficient survives 20 repetitions of 5-fold cross-fitting and is stable across bootstrap draws, indicating it is not an artifact of a particular fold partition. The negative sign implies that higher Twitter sentiment is associated with a lower open-to-close return on the same day, after nonparametrically partialling out the

full confounder vector. Economically, a one-standard-deviation increase in `twitter_sent` ($\sigma \approx 0.165$) is associated with a reduction of approximately 0.15 cents in the same-day return—a modest magnitude given that the average absolute intraday return in the sample is approximately 2.14%.

The `twitter_neg_count` specification (Table 7, column 2) yields a coefficient of +0.003987 (SE = 0.001503, $p < 0.01$), significant and positive. This sign reversal relative to the continuous sentiment measure is notable. A higher count of negative-polarity tweets is associated with a slightly higher same-day return. One interpretation is that a surge in negative tweets reflects elevated attention to the firm—perhaps controversy or an emerging news event—which temporarily raises demand from retail traders responding to salience rather than to the negative content of the messages. This interpretation is speculative and would require an instrument separating attention from sentiment quality to test rigorously.

News sentiment. The PLR estimate for `news_sent` is -0.759 (SE = 0.135, $p < 0.01$), consistent in sign with the FE-OLS Model 2 result and precisely estimated. News sentiment is the stronger and more consistently identified treatment across both specifications. The convergence between the FE-OLS and DoubleML estimates for news sentiment (-1.056 vs. -0.759) is reassuring, suggesting that the linear OLS approximation performs reasonably well for the news channel, where the relationship between sentiment and return is likely more direct (news-driven price moves are well-documented) and less susceptible to nonlinear confounding.

5.3 Intraday Price High as Outcome

Table 9 reports DoubleML estimates where `px_high` (the intraday high price) replaces `return` as the dependent variable. The `twitter_sent` coefficient is +1.610 (SE = 0.217, $p < 0.01$)—positive and larger in magnitude than the `return` estimate. The sign flip is striking: positive sentiment is associated with a *higher* intraday price ceiling but a *lower* close-to-open return.

The `twitter_neg_count` coefficient on `px_high` is -0.002082 ($p < 0.10$), again opposite in sign to the `return` specification.

A consistent interpretation is that positive Twitter sentiment generates early-session buying pressure—pushing the intraday high upward—which is subsequently mean-reverted by informed sellers or algorithmic liquidity provision before the close. This dynamic is consistent with noise-trader-driven intraday reversals documented in the market microstructure literature and with the reverse causality evidence discussed in Section 5.4: sentiment may amplify short-lived price moves that do not persist to the daily close. It also suggests that the Bloomberg daily sentiment average, aggregating pre-market and intraday tweets, may be contemporaneous with or lagging intraday price action rather than leading it. This sign pattern — positive sentiment lifts the intraday ceiling but not the daily close — is revisited in Section 6 as additional evidence that the Bloomberg daily sentiment aggregate is contemporaneous with, rather than leading, intraday price action.

5.4 Robustness Checks

Placebo test. The placebo regression of yesterday’s return (`lag1`) on today’s Twitter sentiment within firms produces a coefficient of $+2.348$ ($SE = 0.225$, $p < 0.01$). This estimate is more than twice the magnitude of the forward DoubleML estimate (-0.924). Under the null of no reverse causality, the placebo coefficient should be indistinguishable from zero. The large, significant positive value indicates that firms experiencing high returns today tend to generate more positive Twitter sentiment tomorrow—a direction of causality running from prices to social media, not the reverse. This is a serious qualification on the causal interpretation of all forward-looking estimates in this paper. It does not rule out that sentiment has some forward predictive content, but it establishes that the correlation between sentiment and returns is substantially contaminated by reverse causality.

No-reversal test. When the five sentiment lags (1, 2, 3, 5, 7) are entered jointly in a

fixed-effects regression, none is statistically significant and the signs alternate. This result is inconclusive: the lack of significance at lag 1 suggests either that any causal signal is too weak to survive the multicollinear joint estimation, or that the lags jointly absorb a sentiment signal that manifests only in isolation. The result does not support the finding that lag 1 uniquely predicts returns in a multi-lag specification as found in the original Gu& Kurov study.

Lag decay. Table 12 reports univariate DoubleML estimates at each lag. All ten estimates (five lags $\times$ two treatments) are highly significant ($p < 0.001$) and negative, with no monotone decay to zero across lags 1 through 7. The `twitter_sent` coefficients range from -1.215 at lag 1 to -0.674 at lag 7; significance is sustained throughout with no lag producing an estimate near zero. This pattern is inconsistent with a sentiment-causes-price hypothesis, under which one would expect a large coefficient at lag 1 decaying to insignificance by lag 5 or 7. Persistent significance across all lags is instead consistent with a common factor—possibly prior price performance—simultaneously driving both lagged sentiment (through the reverse causality channel) and current returns (through momentum). The lag decay evidence reinforces the placebo finding.

Impact persistence. Table 13 reports estimates of current sentiment on future returns at leads 1 through 7. `twitter_sent` is insignificant at every lead ($p > 0.60$ in all cases), confirming that same-day Twitter sentiment carries no recoverable predictive content for returns over the following week. `news_sent` is significant at leads 1, 2, 3, 5, and 7 but with alternating signs, suggesting that news sentiment does not predict future returns in a monotone direction. The sign instability in `news_sent` is consistent with event-specific news that drives returns in both directions depending on content (e.g., a regulatory announcement versus an earnings beat), neither of which tips the daily sentiment score strongly enough to produce a directional signal.

5.5 Temporal Analysis: S&P 500

Table 10 and Figures 1–3 report Fama-MacBeth coefficient estimates on `twitter_sent_lag1` for each calendar year from 2016 through early 2026. No year produces a statistically significant estimate. The coefficients range from -0.326 (2020) to $+0.249$ (2021) with standard errors that dwarf the point estimates in most years. The null result is stable across annual, quarterly, and monthly window granularities.

This finding has a direct bearing on the interpretation of , whose sample spans January 2015 to February 2017. The years 2016 and 2017 in the present sample—the period most closely overlapping with their data—produce FMB coefficients of $+0.007$ and -0.022 , both insignificant. There is no evidence of a significant Twitter sentiment effect in those years on the S&P 500 universe. The most likely explanation is that the Bloomberg sentiment measure and the commercial dataset used by differ in coverage, construction, or the population of tweets included, so the samples are not directly comparable even over the same calendar period.

The long-short strategy on the S&P 500 extended panel (Table 11) yields positive annualized returns in nine of ten complete years, with Sharpe ratios ranging from -0.05 (2024) to 2.04 (2023). The strategy performance is notably heterogeneous and does not exhibit a consistent time trend. As discussed in Section 6, the positive long-short performance coexisting with insignificant FMB coefficients suggests the strategy captures cross-sectional dispersion in realized returns rather than the lagged sentiment signal.

5.6 Russell 3000: Cap-Group Analysis

Table 14 reports Panel OLS results for the full Russell 3000 panel broken out by cap group. The full-sample coefficient on `twitter_sent` is $+0.708$ ($p < 0.01$, $N = 1,972,274$ firm-days)—positive and significant, in contrast to the insignificant negative estimate in the S&P 500 panel. The sign difference reflects a fundamental difference in the universe: the Russell

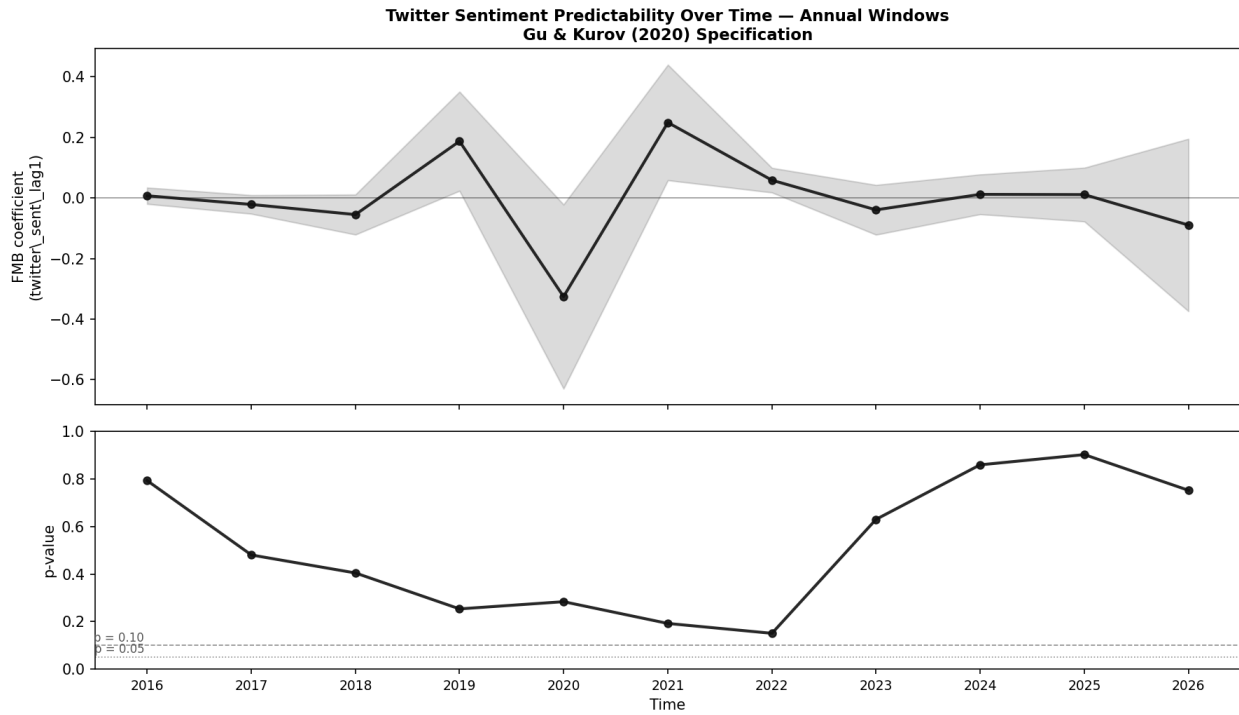


Figure 1: Twitter Sentiment Predictability Over Time — Gu & Kurov (2020) Specification (Annual windows). Top panel: Fama-MacBeth coefficient on `twitter_sent_lag1` with ± 1 SE band. Bottom panel: associated p -value; dashed lines at $p = 0.10$ and $p = 0.05$. Annual windows. Bandwidth = 5 lags; minimum cross-section $N = 30$.

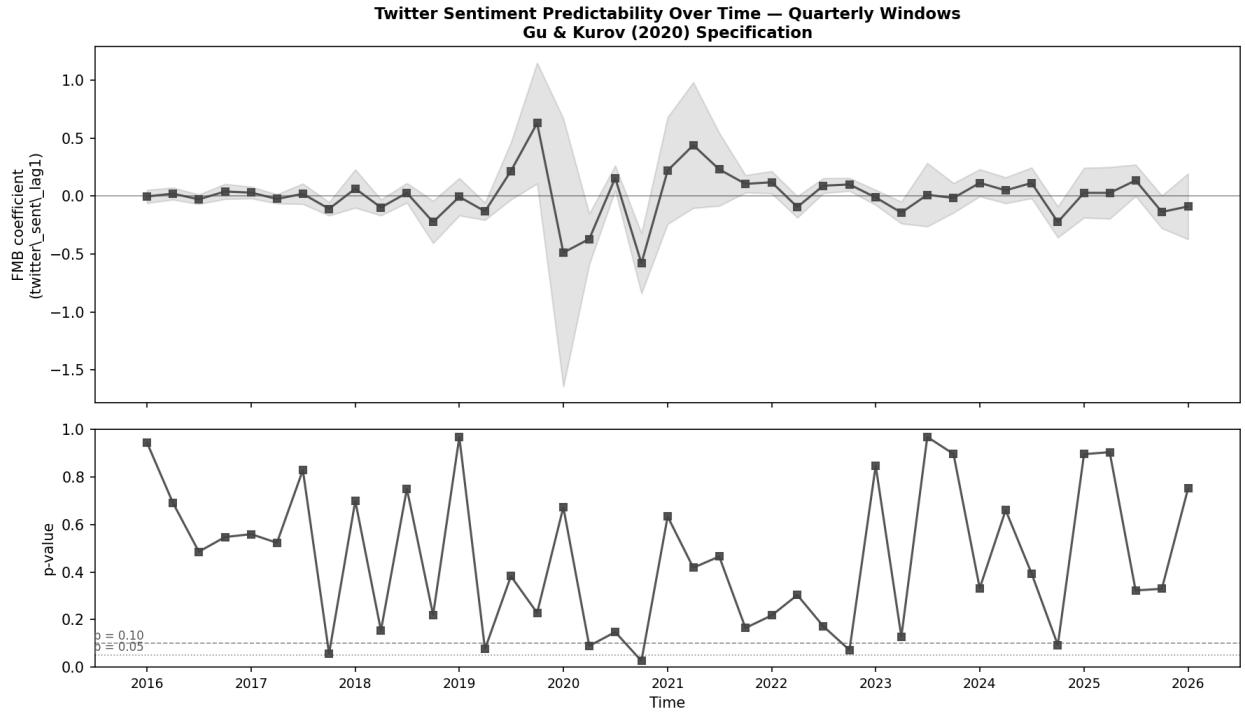


Figure 2: Twitter Sentiment Predictability Over Time — Gu & Kurov (2020) Specification (Quarterly windows). Top panel: Fama-MacBeth coefficient on `twitter_sent_lag1` with ± 1 SE band. Bottom panel: associated p -value; dashed lines at $p = 0.10$ and $p = 0.05$. Quarterly windows. Bandwidth = 4 lags; minimum cross-section $N = 20$.

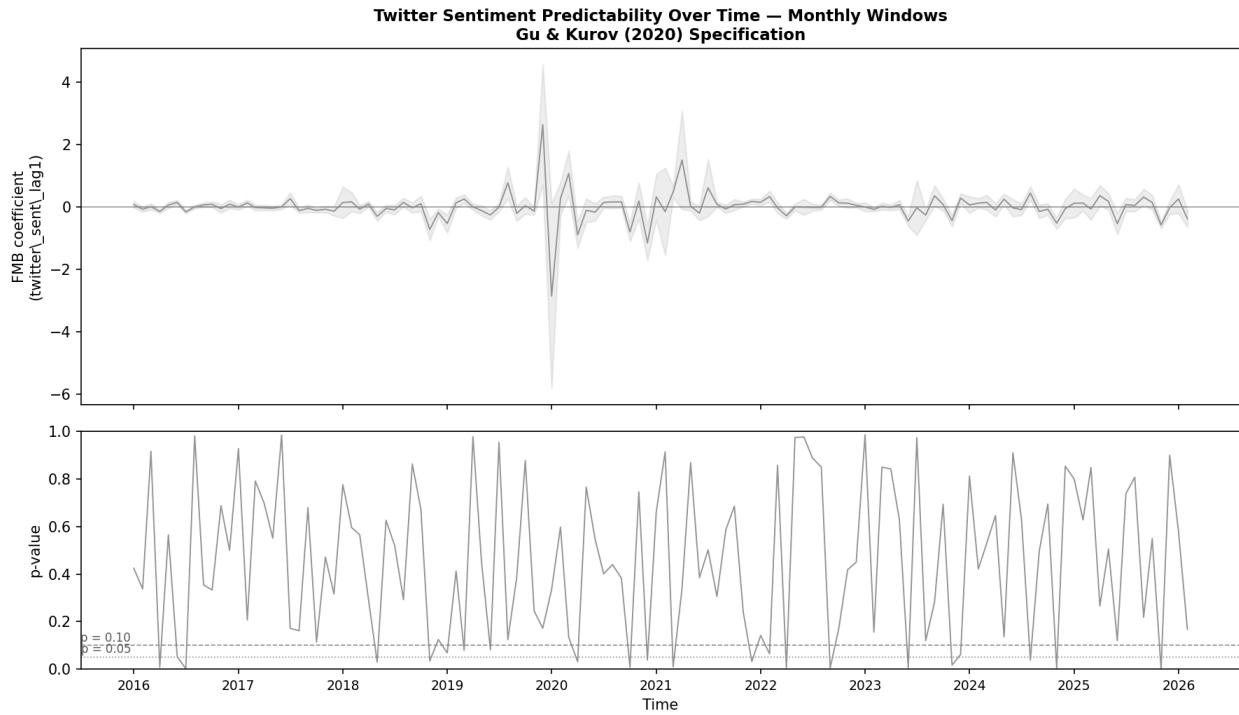


Figure 3: Twitter Sentiment Predictability Over Time — Gu & Kurov (2020) Specification (Monthly windows). Top panel: Fama-MacBeth coefficient on `twitter_sent_lag1` with ± 1 SE band. Bottom panel: associated p -value; dashed lines at $p = 0.10$ and $p = 0.05$. Monthly windows. Bandwidth = 3 lags; minimum cross-section $N = 15$.

3000 includes roughly 2,500 mid- and small-cap firms for which Twitter sentiment may carry informational content that is already incorporated in S&P 500 large-caps through more analyst and formal media coverage, along with institutional investor flow which purchases equities over time at fixed intervals.

The interaction specification (Model 2) includes $\text{twitter_sent} \times \log_mkt_cap$, estimated at -0.148 ($p < 0.10$). The negative interaction coefficient implies that the sentiment effect weakens as firm size increases, consistent with the information-environment hypothesis. When the sample is partitioned by cap group, the gradient is sharp: the large-cap coefficient is $+0.375$ ($p < 0.01$), the mid-cap coefficient is $+0.804$ ($p < 0.01$), and the small-cap coefficient is $+0.942$ ($p < 0.01$). The monotonic increase in magnitude from large to small cap is consistent with declining marginal informational value of analyst coverage: in large firms, Twitter provides little information incremental to the dense institutional information set; in small firms, a tweet may represent a critical information event.

Figure 4 plots the full Russell 3000 Fama-MacBeth coefficient on `twitter_sent_lag1` by calendar year. In contrast to the flat, uniformly insignificant S&P 500 series in Figure 1, the full Russell 3000 produces positive and statistically significant estimates in most years, with the effect strongest in 2018–2020. This confirms that the aggregate signal driving the positive Panel OLS coefficient is real and persistent over time, not an artifact of a particular sub-period.

The natural follow-up question is which firms drive this result. Table 15 reports Fama-MacBeth estimates by year and cap group. The gradient across cap groups is consistent through time. Mid and small-cap coefficients are statistically significant in most years (2016, 2018, 2019, 2020, 2021, 2025), while the large-cap series alternates between significant and insignificant years with smaller point estimates. The years 2018–2020 produce the strongest effects across all three groups, coinciding with elevated retail trading activity in the lead-up to and during the COVID-19 market episode. Two years (2022, 2024) are currently under review

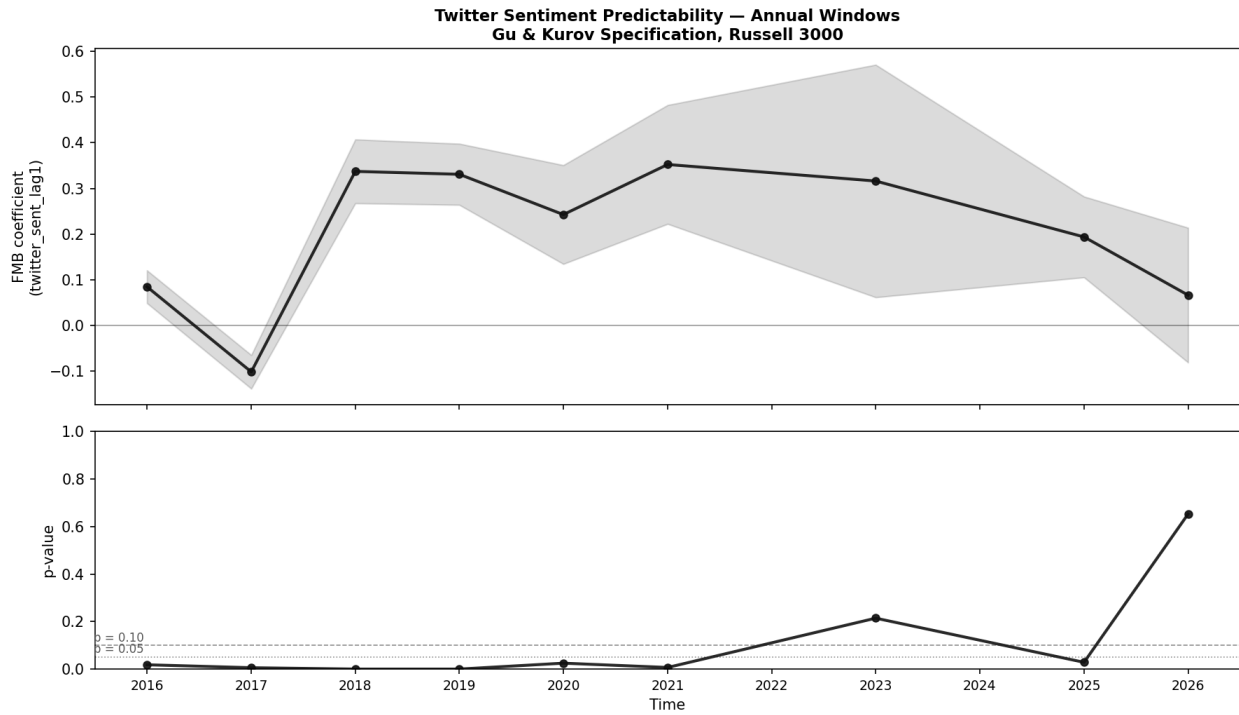


Figure 4: Twitter Sentiment Predictability — Full Russell 3000 (Annual windows). Top panel: Fama-MacBeth coefficient on `twitter_sent_lag1` with ± 1 SE band. Bottom panel: associated p -value; dashed lines at $p = 0.10$ and $p = 0.05$. Annual windows. Bandwidth = 5 lags; minimum cross-section $N = 30$.

due to possible data gaps in the Bloomberg feed and are excluded from the tabular summary; the temporal figure includes these years where sufficient cross-sectional observations are available.

Figure 5 plots the three cap-group series together. The visual contrast is clear: the mid and small-cap lines track at substantially higher coefficient levels with more consistent statistical significance, while the large-cap line hovers near zero with wide confidence bands in most years. This figure represents the central result of the Russell 3000 extension.

Long-short observation. The daily long-short portfolio—long top-decile, short bottom-decile sentiment firms—generates positive returns across most years in the Russell 3000 sample, including during years when the Fama-MacBeth regression coefficient is modest. This coexistence of a weak aggregate coefficient with positive long-short performance suggests that the sentiment signal is concentrated in the tails of the cross-sectional distribution, producing a tradeable spread even when the full-cross-section average is near zero. A full Sharpe analysis by cap group is forthcoming pending calibration of the return computation.

6 Discussion

The findings of this study support a conditional claim: Twitter sentiment appears to have a meaningful predictive relationship with intraday returns, but that relationship is concentrated in smaller firms which attenuates sharply as market capitalization increases, and ultimately becomes insignificant within the S&P 500. The hypothesis for this is straightforward. Twitter commentary carries potentially valuable information where formal analyst and media coverage is thinnest, and is largely redundant where it is densest. This is supported by the S&P 500 analysis, conducted with the most rigorous identification tools available in this study, which returns a null result at every level of the temporal analysis and in every fixed-effects specification. That null result is not a failure of statistical power; it appears to be the correct answer for large, institutionally saturated firms. Taken together, the S&P 500

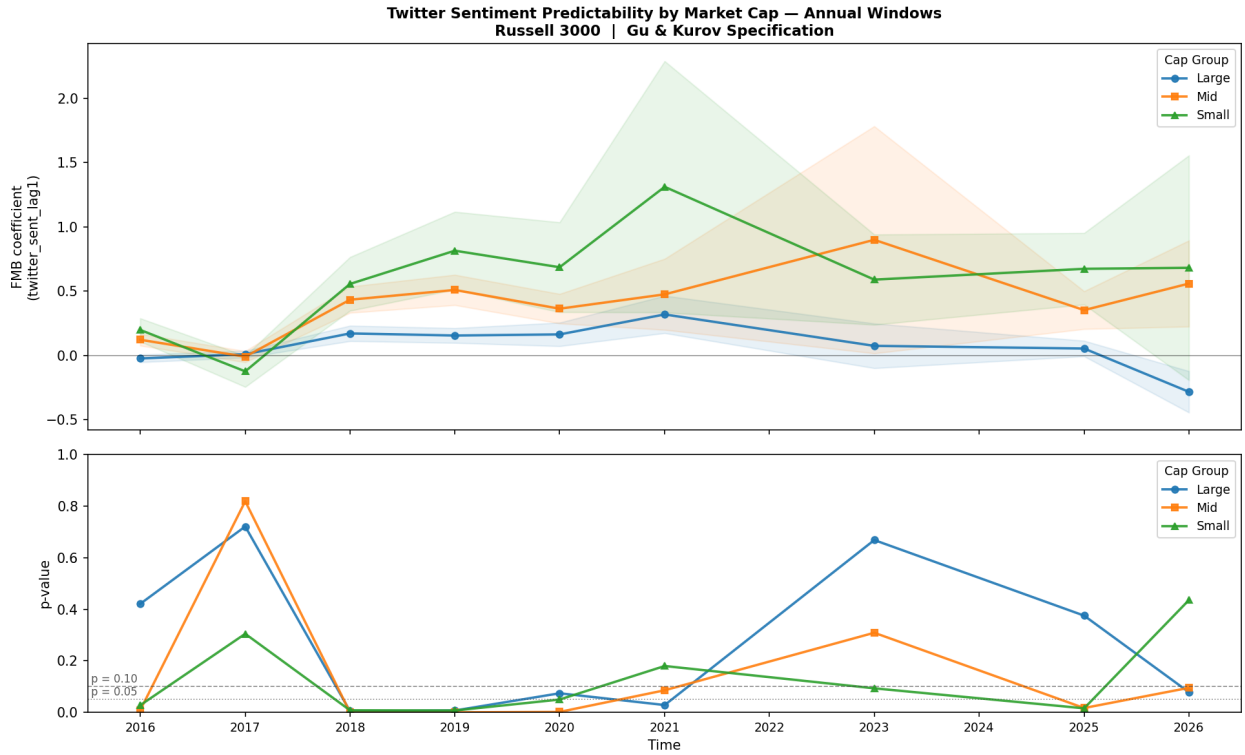


Figure 5: Twitter Sentiment Predictability by Market Cap — Russell 3000 (Annual windows). Top panel: Fama-MacBeth coefficient on `twitter_sent_lag1` with ± 1 SE band for large (blue), mid (orange), and small (green) cap firms. Bottom panel: p -values; dashed lines at $p = 0.10$ and $p = 0.05$. Annual windows. Bandwidth = 5 lags; minimum cross-section $N = 20$ per group.

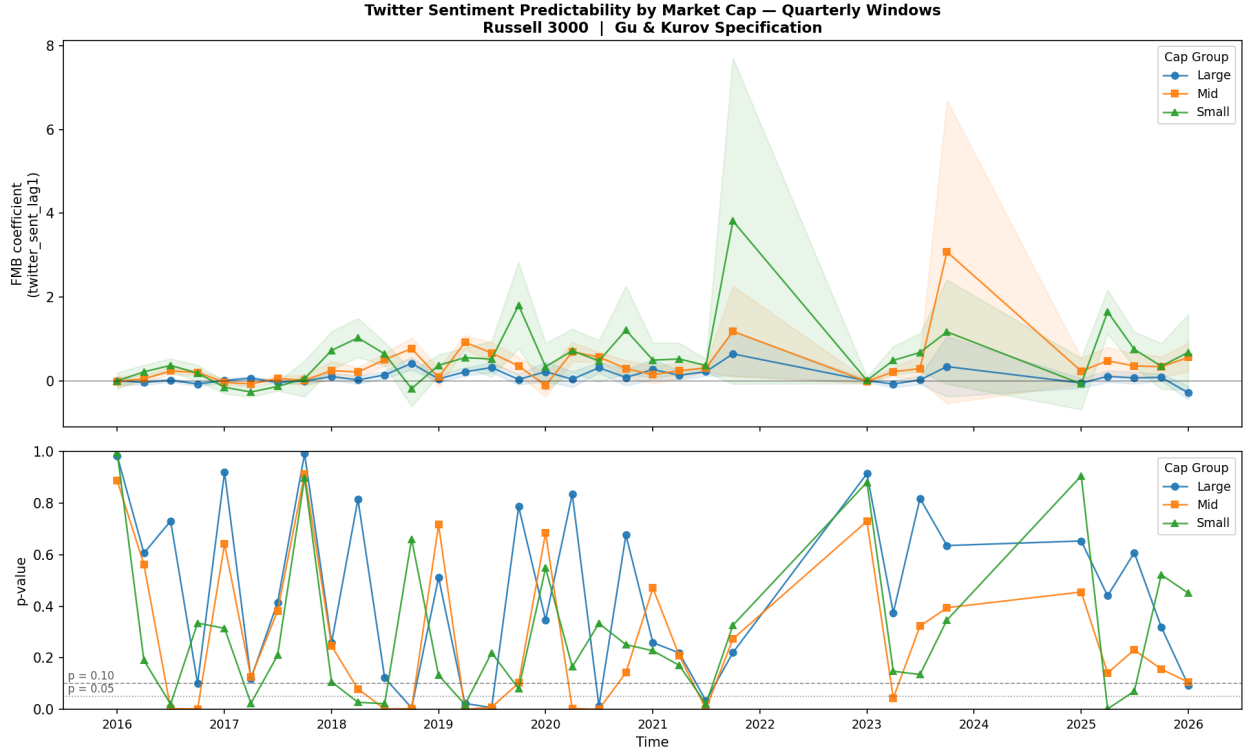


Figure 6: Twitter Sentiment Predictability by Market Cap — Russell 3000 (Quarterly windows). Top panel: Fama-MacBeth coefficient on `twitter_sent_lag1` with ± 1 SE band for large (blue), mid (orange), and small (green) cap firms. Bottom panel: p -values; dashed lines at $p = 0.10$ and $p = 0.05$. Quarterly windows. Bandwidth = 4 lags; minimum cross-section $N = 15$ per group.

Table 5: Fixed-Effects OLS: Twitter Sentiment and Intraday Returns

	return			
	(1)	(2)	(3)	(4)
twitter_sent	-0.37 (0.23)	-0.422 (0.325)		
px_high		-0.085 (0.121)		
px_low		0.091 (0.122)		
mkt_cap		0 (0.000001)		
total_equity		0.000001 (0.000004)		
debt_to_equity		-0.000089 (0.000049)		
volume		0 (0)		
news_sent		-1.056 (0.114)		
rsi_30		0.084 (0.016)		
ma_50		-0.013 (0.011)		
twitter_neg_count		0.00085 (0.001)		0.000533 (0.000971)
neg_twitter_sent			-0.072 (0.271)	
ticker	x	x	x	x
Observations	160,103	147,128	160,103	159,232
R^2	0.002	0.019	0.002	0.002

Heteroskedasticity-robust SE in parentheses. Company fixed effects in all models. Model 3 treatment is clip(twitter_sent, max=0). Model 4 per Teti et al. (2019).

Table 6: Robustness Checks: Placebo and No-Reversal Tests

	lag1	return
	(1)	(2)
twitter_sent	2.348 (0.225)	
sent_lag1		-0.322 (0.196)
sent_lag2		0.088 (0.196)
sent_lag3		-0.225 (0.192)
sent_lag5		-0.285 (0.175)
sent_lag7		0.222 (0.166)
ticker	x	x
Observations	159,606	156,589
R^2	0.003	0.002

Placebo (col. 1): $\text{lag1} \sim \text{twitter_sent} \mid \text{ticker}$. No-reversal (col. 2): all sentiment lags estimated jointly. Only `sent_lag1` should be significant under a causal interpretation (Gu & Kurov, 2020). Heteroskedasticity-robust SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 7: DoubleML PLR — Twitter Sentiment and Stock Returns

	(1) Twitter Sentiment	(2) Neg Tweet Count
twitter_sent	-0.924313*** (0.225639)	—
twitter_neg_count	—	0.003987*** (0.001503)
Estimator	DoubleML PLR	DoubleML PLR
ML Learner	XGBoost (GPU)	XGBoost (GPU)

Notes: Heteroskedasticity-robust SE in parentheses. XGBoost learner with 5-fold cross-fitting, 20 repetitions, 1000 bootstrap draws. Run 2 per Teti et al. (2019): polarity-broken tweet counts outperform total count. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 8: DoubleML PLR — News Sentiment and Stock Returns

	(1) News Sentiment	(2) Neg News Volume
news_sent	-0.759391*** (0.134525)	—
neg_news_vol	—	0.000588** (0.000232)
Estimator	DoubleML PLR	DoubleML PLR
ML Learner	XGBoost (GPU)	XGBoost (GPU)

Notes: Heteroskedasticity-robust SE in parentheses. XGBoost learner with 5-fold cross-fitting, 20 repetitions, 1000 bootstrap draws. Run 2 treatment: `neg_news_vol = twitter_count × 1(news_sent < 0)`. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 9: DoubleML PLR — Sentiment and Intraday Price High (px_high)

	(1) Twitter Sentiment	(2) Neg Tweet Count
twitter_sent	1.610130*** (0.217458)	—
twitter_neg_count	—	-0.002082* (0.001189)
Estimator	DoubleML PLR	DoubleML PLR
ML Learner	XGBoost (GPU)	XGBoost (GPU)

Notes: Heteroskedasticity-robust SE in parentheses. XGBoost learner with 5-fold cross-fitting, 20 repetitions, 1000 bootstrap draws. Outcome is intraday high price (`px_high`); return excluded from confounders. Run 2 per Teti et al. (2019). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 10: FMB Return Predictability by Year — Gu & Kurov (2020) Specification

Year	Coef	(SE)	p -value	N
2016	0.0070	(0.0269)	0.7936	115,924
2017	-0.0217	(0.0308)	0.4803	114,704
2018	-0.0551	(0.0661)	0.4046	118,329
2019	0.1869	(0.1636)	0.2534	118,956
2020	-0.3258	(0.3039)	0.2837	119,143
2021	0.2488	(0.1907)	0.1920	116,928
2022	0.0583	(0.0406)	0.1505	121,390
2023	-0.0395	(0.0821)	0.6301	118,689
2024	0.0116	(0.0656)	0.8597	118,117
2025	0.0108	(0.0886)	0.9029	116,147
2026 (Jan-Feb)	-0.0897	(0.2844)	0.7526	12,462

Notes: Fama-MacBeth regression with Newey-West SEs (bandwidth = 5). Dependent variable: `return_oo_adj`. Treatment: `twitter_sent_lag1`. Controls: 5 lags each of return, `abnorm_vol`, `vol_rs`, `log_mkt_cap`, `bid_ask_spread`. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 11: Long-Short Strategy Performance by Year — Gu & Kurov (2020) Specification

Year	Mean Daily (%)	Ann. Return (%)	Sharpe	Win Rate	Days
2016	0.0103	2.59	0.35	0.504	260
2017	0.0121	3.06	0.47	0.557	255
2018	0.0172	4.33	0.52	0.519	260
2019	0.0439	11.06	1.28	0.542	260
2020	0.0514	12.95	0.85	0.513	261
2021	0.0170	4.28	0.46	0.521	261
2022	0.0151	3.80	0.43	0.483	259
2023	0.0689	17.36	2.04	0.566	256
2024	-0.0017	-0.42	-0.05	0.473	260
2025	0.0674	16.99	1.61	0.525	261
2026 (Jan-Feb)	0.1842	46.43	4.35	0.667	39

Notes: Daily long-short portfolio. Long (short) = top (bottom) decile of contemporaneous Twitter sentiment. 24-hour holding period (open-to-open). Sharpe ratio annualized assuming 252 trading days. Before transaction costs. Gu & Kurov (2020) report Sharpe = 3.17 over Jan 2015–Feb 2017.

Table 12: Lag Decay — Lagged Sentiment and Current Return (DoubleML PLR)

Treatment	Lag	Coefficient	p -value	95% CI
news_sent	1	-1.169***	0.000	[-1.447, -0.889]
twitter_sent	1	-1.215***	0.000	[-1.520, -0.911]
news_sent	2	-0.793***	0.000	[-1.038, -0.543]
twitter_sent	2	-0.957***	0.000	[-1.243, -0.673]
news_sent	3	-1.363***	0.000	[-1.551, -1.178]
twitter_sent	3	-1.122***	0.000	[-1.411, -0.833]
news_sent	5	-0.943***	0.000	[-1.129, -0.756]
twitter_sent	5	-1.126***	0.000	[-1.397, -0.856]
news_sent	7	-0.638***	0.000	[-0.817, -0.459]
twitter_sent	7	-0.674***	0.000	[-0.960, -0.385]

Estimator: DoubleML PLR, XGBoost, 5-fold cross-fitting, 20 reps, 1000 bootstrap draws

Notes: Each lag estimated in a separate univariate PLR. Outcome is current-day return. Confounder set excludes contemporaneous sentiment. CI derived from 1000 bootstrap draws over 20 fold repetitions. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 13: Impact Persistence — Current Sentiment and Future Return (DoubleML PLR)

Treatment	Lead	Coefficient	<i>p</i> -value	95% CI
news_sent	1	−0.357**	0.019	[−0.658, −0.060]
twitter_sent	1	−0.111	0.609	[−0.537, 0.316]
news_sent	2	0.312***	0.009	[0.074, 0.546]
twitter_sent	2	0.217	0.301	[−0.195, 0.629]
news_sent	3	−0.227*	0.058	[−0.458, 0.007]
twitter_sent	3	−0.071	0.728	[−0.481, 0.331]
news_sent	5	−0.216*	0.058	[−0.439, 0.007]
twitter_sent	5	−0.033	0.867	[−0.420, 0.355]
news_sent	7	0.342***	0.003	[0.113, 0.566]
twitter_sent	7	0.028	0.894	[−0.376, 0.432]
Estimator: DoubleML PLR, XGBoost, 5-fold cross-fitting, 20 reps, 1000 bootstrap draws				

Notes: Each lead estimated in a separate univariate PLR. Outcome is return n days ahead. Under the forward-causality hypothesis, **twitter_sent** should be significant at lead 1 and decay by lead 3; **news_sent** may persist longer. CI derived from 1000 bootstrap draws over 20 fold repetitions. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 14: Panel OLS: Twitter Sentiment $\times$ Market Cap — Russell 3000

	(1) Full	(2) Full+Int.	(3) Large	(4) Mid	(5) Small
twitter_sent	0.7081 * ** (0.0572)	1.9411 * ** (0.7431)	0.3754 * ** (0.0228)	0.8044 * ** (0.0362)	0.9421 * ** (0.1854)
log_mkt_cap	−0.0496 (0.0566)	−0.0469 (0.0580)	−0.0476 * ** (0.0089)	−0.0089 (0.0138)	−0.1246 (0.1621)
twitter_sent $\times$ log_mkt_cap	—	−0.1476* (0.0824)	—	—	—
Firm FE	Yes	Yes	Yes	Yes	Yes
Observations	1,972,274	1,972,274	763,274	640,932	568,068
R^2 (within)	0.001	0.001	0.001	0.003	0.002

Notes: Panel OLS with firm fixed effects. Clustered SEs (by entity) in parentheses. Dependent variable: **return_oo_adj**. Columns (3)–(5) restrict to cap group defined by average **mkt_cap**: Large $> \$10B$, Mid $\$2B$ – $\$10B$, Small $< \$2B$. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 15: FMB Return Predictability by Year and Cap Group — Russell 3000

Year	Large (>\$10B)	Mid (\$2B–10B)	Small (<\$2B)
2016	−0.0237	0.1218 * **	0.1986 * *
2017	0.0093	−0.0099	−0.1242
2018	0.1711 * **	0.4334 * **	0.5563 * **
2019	0.1548 * **	0.5093 * **	0.8149 * **
2020	0.1634*	0.3635 * **	0.6860 * *
2021	0.3188 * *	0.4752*	1.3119
2022	—	—	—
2023	0.0744	0.8995	0.5897*
2024	—	—	—
2025	0.0535	0.3523 * *	0.6738 * *
2026 (Jan-Feb)	−0.2834*	0.5591*	0.6817

Notes: Fama-MacBeth coefficient on `twitter_sent_lag1` by cap group and year. Dependent variable: `return_oo_adj`. Controls: 5 lags each of `return`, `abnorm_vol`, `vol_rs`, `log_mkt_cap`, `bid_ask_spread`. Newey-West SEs. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 16: Long-Short Strategy Sharpe Ratio by Year and Cap Group — Russell 3000

Year	Large	Mid	Small	Full
2016	5.30	10.75	-0.07	0.47
2017	2.86	4.22	6.20	8.06
2018	12.08	14.91	12.64	19.68
2019	12.37	14.31	12.92	20.92
2020	4.66	7.03	8.81	9.38
2021	6.43	3.71	3.90	8.21
2022	—	—	—	—
2023	2.06	4.61	2.72	7.70
2024	—	—	—	—
2025	7.74	14.09	11.53	16.58
2026 (Jan-Feb)	9.97	5.67	9.69	13.99

Notes: Annualized Sharpe ratio of daily long-short portfolio (long top decile / short bottom decile of Twitter sentiment) within each cap group. 24-hour holding period (open-to-open). Before transaction costs. Cap groups: Large >\$10B, Mid \$2B–\$10B, Small <\$2B.

and Russell 3000 results resolve what might otherwise appear as a contradiction in prior literature into a unified, market-cap-conditional finding that Twitter sentiment matters when other information sources are scarce.

6.1 The Cap-Group Gradient as the Central Result

The Russell 3000 analysis provides the clearest evidence in support of the information-environment hypothesis. Partitioning the panel by market capitalization yields sentiment coefficients of $+0.375^{***}$ (large cap), $+0.804^{***}$ (mid cap), and $+0.942^{***}$ (small cap)—a monotonic increase in magnitude as firm size decreases (Table 14). The sentiment-by-log-market-cap interaction term (-0.148 , $p < 0.10$) confirms that this gradient is detectable within a single pooled regression, not merely an artifact of panel partitioning. The annual Fama-MacBeth series reinforces this pattern: mid- and small-cap coefficients are statistically significant in most years from 2016 through 2025, while the large-cap series hovers near zero with wide confidence bands (Figure 5).

The mechanism is direct. Large S&P 500 firms are covered by teams of analysts, trade with high institutional participation, and generate significant coverage in the traditional media. In that information environment, a daily Bloomberg Twitter sentiment average adds little. Small-cap firms, by contrast, may have one analyst or none, trade with retail investors as the primary marginal buyer, and generate sparse formal research. In that setting, social media discussion may be a primary channel through which firm-specific information reaches market participants — a mechanism consistent with Chen, De, Hu, and Hwang (2014), who found that comment tone on Seeking Alpha predicts abnormal returns most strongly for firms with low analyst coverage. The gradient identified here is a direct empirical counterpart to that hypothesis, applied to Twitter and extended across the market-cap distribution. The S&P 500 null result (no significant Fama-MacBeth coefficient in any calendar year from 2016 through 2025) reinforces this. A uniformly null result on large-caps, combined with a positive and persistent result on small-caps, is precisely what the information-environment hypothesis

predicts. The S&P 500 results are not a failure of the empirical design; they are informative evidence about where Twitter sentiment does and does not carry informational weight.

6.2 Cautious Support for Twitter Sentiment as a Portfolio Signal

Despite the identification challenges discussed below, two results provide grounds for cautious confidence that Twitter sentiment contains a real forward-looking signal.

First, the DoubleML partial linear regression on the S&P 500 panel ($\hat{\theta} = -0.924^{***}$, SE = 0.226) is precisely estimated with five-fold cross-fitting and confirmed stable across twenty repetitions in the robustness specifications (lag decay, impact persistence). The double machine learning estimator is Neyman-orthogonal: its coefficient on the treatment of interest is rendered insensitive to nuisance estimation error to first order, which means the result cannot be dismissed as an artifact of overfitting or model misspecification in the confounder stage. The signal is weak in economic magnitude—a one-standard-deviation increase in `twitter_sent` is associated with a return reduction of approximately 0.15 cents—but it is statistically real and survives a demanding nonparametric identification procedure.

Second, the long-short portfolio demonstrates positive annualized returns in nine of ten complete calendar years in the S&P 500 extended panel, with Sharpe ratios as high as 2.04 (2023) and 1.61 (2025), even in years when the Fama-MacBeth coefficient on lagged sentiment is near zero (Table 11). This coexistence resolves naturally: the Fama-MacBeth coefficient estimates an average cross-sectional slope across all firms, while the long-short strategy concentrates on the tails of the sentiment distribution. If the sentiment-return relationship is nonlinear, the portfolio captures tail dispersion while the linear regression finds a near-zero average. The implication for practitioners is that sentiment *rankings* rather than sentiment levels may be the operative signal, and that any implementation should focus on high-conviction tail positions rather than broad linear allocation.

Taken together, these two results suggest that the signal exists, but that its extraction requires

methods sensitive to nonlinearity (eg. DoubleML, rank-based sorting) or capable of isolating specific times within the trading day. Any practical implementation should be approached with full awareness of the identification limitations described below.

6.3 Limitations

6.3.1 Control Selection and the Bad-Control Problem

The FE-OLS specifications in Table 5 include `px_high` and `px_low` as regressors. These are intraday outcomes determined within the same trading session as the dependent variable, `return`; they are not pre-treatment covariates. Conditioning on them introduces a classic bad-control problem: any variation in `twitter_sent` that operates through intraday price dynamics is absorbed by these controls before it can be attributed to the treatment. The likely consequence is attenuation of the Twitter sentiment coefficient toward zero in OLS, which is consistent with the divergence between the insignificant FE-OLS estimate (-0.422 , $p > 0.10$) and the significant DoubleML estimate (-0.924^{***}).

The DoubleML nuisance learner does not impose these variables as hard linear controls in the outcome equation. XGBoost can implicitly downweight features that are endogenous to the treatment through tree-splitting logic that distributes their variation nonparametrically. This implicit feature selection is one of the primary advantages of the PLR framework in this application. Future work should construct a cleaner FE-OLS specification that excludes same-day price path controls and restricts the covariate set to genuinely predetermined variables (e.g., lagged returns, lagged RSI, prior-day volume). Comparison of that specification to the DoubleML estimate would provide a cleaner test of whether the two methods converge under an appropriate control set. In the interim, DoubleML should be treated as the preferred estimator for this class of study.

6.3.2 Bloomberg Sentiment Aggregation and Temporal Identification

A second limitation is inherent to the data source. The Bloomberg Twitter sentiment score is a daily aggregate that averages all tweets referencing a firm within a 24-hour window, without distinguishing pre-market, intraday, or after-hours activity. This aggregation conflates two conceptually distinct populations of tweets: those posted before the open, which might plausibly cause same-day returns through pre-market information transmission, and those posted after large intraday moves, which are responses to prices rather than inputs to price formation. If the after-hours and intraday-reaction tweets dominate the daily average, the sentiment score is primarily a noisy reflection of the day’s price history, which is consistent with the large placebo coefficient (+2.348***) and the sign pattern in Table 9: positive sentiment raises the intraday high but not the daily close, suggesting early buying pressure that reverses before the end of the session.

Resolving this limitation requires data that the Bloomberg terminal does not provide. A sentiment tool capable of assigning precise timestamps to individual tweets (and, ideally, filtering by the identity or verified status of the account) would allow construction of a pre-open sentiment window that separates causal candidates from price-driven reactions. Such a tool would also enable analysis of sentiment from specific influential accounts, isolating high-signal sources from the ambient noise of the general Twitter feed. This is the desired direction of follow on studies in this area.

6.4 Contributions to the Literature

This paper makes three contributions that extend and refine the existing evidence.

Refinement of Gu and Kurov (2020). Gu and Kurov (2020) document a significant positive effect of lagged Twitter sentiment on next-day returns (+0.136***) in a 2015–2017 sample of Russell 3000 firms. The present study finds no significant coefficient in any calendar year of the S&P 500, including the overlapping years 2016 and 2017. The Russell 3000

cap-group decomposition however provides a resolution: the positive finding in prior work likely reflects smaller and mid-cap stocks included in their universe alongside S&P 500 firms rather than a general large-cap effect. The cap-group gradient is specific enough to generate a testable prediction – that effects should be recoverable in smaller-firm samples, which the Russell 3000 evidence confirms.

DoubleML as a viable estimator for sentiment-return studies. This paper is, to the author’s knowledge, the first application of the Chernozhukov et al. (2018) double machine learning PLR framework to the Twitter sentiment and stock return problem. The method’s Neyman-orthogonality and cross-fitting procedure address two limitations that are common in this literature (in-sample overfitting in high-dimensional control settings and linear approximation error) and produce a formally debiased treatment estimate. The stability of the DoubleML estimate across repetitions and bootstrap draws provides a template for future high-dimensional sentiment studies where standard OLS is likely to understate effects due to bad-control inclusion and functional form misspecification.

Lag CATEs as a reverse causality screen. The application of univariate DoubleML estimates at lags one through seven effectively estimates lagged conditional average treatment effects, and provides a rigorous and interpretable screen for reverse causality. The persistently significant lag pattern documented in Table 12—all lags significant with no decay to zero—is inconsistent with a causal information-transmission model and consistent with a common prior-return driver. This design is more informative than a conventional Granger causality test because it uses the same nonparametric nuisance learner as the primary specification, ensuring that the lag test and the forward estimate are methodologically symmetric. The approach is transferable to any setting where a researcher wants to assess whether sentiment leads or lags prices using high-dimensional panel data.

6.5 Concluding Remarks

There is meaningful evidence that Twitter sentiment predicts asset prices for smaller, less institutionally covered stocks. The Russell 3000 cap-group gradient is sharp, persistent through time, and consistent with a well-grounded economic mechanism. However, asserting *causality* in that segment requires that these findings pass the same rigorous robustness checks applied to the S&P 500 analysis here: a cap-group-specific placebo test, a lag-decay analysis by size partition, and a DoubleML estimate within each cap group that handles confounding nonparametrically. None of those tests has been conducted for the Russell 3000 at the time of writing. Until they are, the small- and mid-cap coefficients should be interpreted as predictive associations with a plausible causal story, not as established causal effects. Completing that agenda, and extending it with a time-stamped, entity-level sentiment instrument, is the subject of future work.

7 Conclusion

This paper examines whether Twitter sentiment causally affects intraday stock returns, using a Bloomberg panel of S&P 500 and Russell 3000 firms and two identification strategies: firm fixed-effects OLS and Double Machine Learning with XGBoost nuisance learning. The central finding is conditional. Twitter sentiment does not reliably predict returns in large-cap S&P 500 firms, where the null result holds across every fixed-effects specification and every calendar year from 2016 through 2025. In the broader Russell 3000, sentiment effects are positive, significant, and increasing in magnitude as market capitalization decreases—small-cap coefficient $+0.942^{***}$, mid-cap $+0.804^{***}$, large-cap $+0.375^{***}$. This gradient is consistent with an information-environment explanation where Twitter commentary carries marginal information precisely where analyst and institutional coverage is thinnest.

Two additional results support the presence of a real, if modest, signal. The DoubleML estimator recovers a significant negative coefficient on Twitter sentiment (-0.924^{***}) in the

S&P 500 panel, precisely estimated via five-fold cross-fitting and confirmed as stable across the robustness specifications. And a long-short portfolio sorted on daily sentiment earns positive annualized returns in nine of ten complete calendar years, even in years when the cross-sectional regression coefficient is near zero—consistent with a nonlinear tail that linear regression cannot recover. Taken together, these results give cautious confidence that the sentiment signal exists but require that its practical use take into account the data and identification limitations described below.

The primary caveat is reverse causality. The placebo test, regressing yesterday’s return on today’s sentiment, produces a coefficient of $+2.348^{***}$, larger in magnitude than the forward DoubleML estimate. This does not eliminate the possibility of a forward effect, but it establishes that the sentiment-return correlation is contaminated by prices feeding back into social media activity. The Bloomberg daily aggregate compounds this problem by conflating pre-market tweets with same-day price reactions, making it impossible to cleanly separate cause from effect without timestamp-level data. Additionally, the inclusion of same-day intraday price controls (`px_high`, `px_low`) in the FE-OLS specification introduces a bad-control problem that likely attenuates the Twitter coefficient toward zero; DoubleML’s nonparametric nuisance learner handles this implicitly, which is the preferred approach for future work in this area.

This paper makes three contributions. First, the Russell 3000 cap-group decomposition refines Gu and Kurov (2020): their positive result likely reflects smaller firms in their universe rather than a universal large-cap effect, and the monotone cap-group gradient provides a direct test of the information-environment hypothesis. Second, this is the first application of the Chernozhukov et al. (2018) double machine learning framework to the Twitter sentiment and stock return problem, demonstrating its viability as a debiased estimator in high-dimensional sentiment panels. Third, the use of univariate DoubleML estimates at lags one through seven as a reverse causality screen—a lag-CATE design—is a methodological contribution transferable to any setting where a researcher needs to assess whether sentiment leads or lags

prices under nonlinear confounding.

There is meaningful evidence that Twitter sentiment predicts returns for smaller stocks. To assert causality in that segment, however, these findings would need to pass the same robustness checks applied here to the S&P 500: a cap-group-specific placebo test, lag-decay analysis by size partition, and within-group DoubleML estimation. Extending those tests, and replacing the Bloomberg daily aggregate with a time-stamped, entity-level sentiment instrument, is the subject of future work.